

## THE FIRST TWO STEPS AFTER THE SIGNAL ARRIVES: VERIFY AND TELL THE WORLD

PETER B. BOYCE

American Astronomical Society, 2000 Florida Avenue, Suite 300, Washington, DC 20009 USA

**Abstract**—The Discovery and subsequent verification of a suspected ETI signal will have the most profound impact imaginable upon all inhabitants of our planet. The first responsibility of the community involved in SETI is to assure that a suspected ETI signal is real. The second responsibility is to guarantee that the signal is subjected to objective, scientific analysis.

### 1. INTRODUCTION

To assure the reality of a suspected signal, one recommendation, which has been well received by many SETI workers, is to require confirmation by independent observations at a second site. Without this confirmation, no suspected signal will be accepted as real.

To guarantee that a confirmed signal will be subjected to objective analysis, it is important that accurate, specific information about the signal be spread to scientists around the world in a free and open manner. Many years ago the astronomical community developed a method for rapid communication about unexpected events. As an example, news about the supernova 1987A was sent to the International Astronomical Union Telegram Bureau immediately. The supernova was confirmed shortly, and specific scientific observations were under way at several observatories within hours. The International Ultraviolet Explorer satellite started observing the supernova within three hours of its discovery. The IAU system of telegrams works very well.

Because this is the first naked-eye supernova in nearly 400 years, there is great public interest. This situation is exactly that which will occur when and ETI signal is first detected. The SETI community should agree to use the same time-tested, well-established methods to spread the information about ETI signals. By utilizing these communication channels which go from scientist to scientist and which will be periodically tested by the occurrence of unusual astronomical events, we will guarantee that an ETI signal will be shared by the whole world in a rapid and effective fashion.

### 2. BACKGROUND

Discovery of an ETI signal would be of great importance to the whole world. The comments by Hesburgh[1] and Morrison[2] eloquently testify to this. Such a discovery would affect

every country and would spark intense interest on the part of nearly every civilized person. In addition to being a momentous discovery in its own right, the discovery of another civilization within our galaxy will have far reaching effects upon our perspective of the universe. Any suspected ETI signal must be verified as rapidly as possible.

As was pointed out by the author[3] in 1986, it may be a very difficult job to verify the extraterrestrial origin of such a signal. However, the issue is so important and the possibility of spurious terrestrial signals being taken as an ETI signal is sufficiently high that verification by all possible methods should be sought. Since there is no way of knowing how long an ETI signal may last, it is vital to start the confirmation process as rapidly as possible. To do this it will be necessary to disseminate information about a suspected ETI signal. After verification the job becomes one of getting accurate information to the public.

### 3. TYPES OF DISSEMINATION

#### 3.1 To scientist - for verification

By far the largest efforts at detecting ETI signals are being carried out using the techniques and instruments of radio astronomy. Astronomers using other regions of the electromagnetic spectrum such as the infrared, are also aware of the possibilities that unusual signals may have an ETI origin, although I believe it is likely that the first ETI signal will be detected in the microwave region of the radio spectrum. The conclusions and recommendations presented in this paper are independent of what frequency the signal comes in on.

Assuming a radio signal is detected, and passes all the tests for ETI origin which can be made by the discovery instrument, the next step is to let other scientists know the details so they can confirm the discovery. For several reasons radio astronomers are in the best position to

confirm such a signal. Powerful radio telescopes are in daily use at a number of sites throughout the world. The scheduling of all astronomical telescopes is very flexible. A target of this importance can often be acquired and observed within minutes of notification. In addition, astronomical telescopes are equipped with flexible and adaptable instrumentation, which can be set up to match any unusual characteristics of an ETI signal. Finally, radio astronomers have the possibility of using interferometry techniques in order to pinpoint the position of a signal and provide the most powerful technique for eliminating confusion by stray terrestrial signals and deliberate hoaxes. Therefore, the first information about a suspected ETI signal and a request for verification will have to be directed to radio astronomers around the world.

### 3.2 To the Public - After Initial Verification

However, we cannot and should not stop after notifying fellow scientists of the SETI community. The whole world will want to know about an ETI signal. With the example before us of the spread of information about the recent nearby supernova, SN1987A, we can be assured that the news media will take care of spreading the word to the general public. As scientists, we in the SETI community will have the responsibility to see that factual information and informed commentary get to the news media. This can be difficult.

In cases where there is an immense public interest, news reporters, who are not familiar with science issues, will be reporting the story. Such reporters will often have an imperfect understanding of the situation, or be working with a garbled version of the facts. They will, however, come to local scientists for information or comments. Unfortunately, the scientists will often lack up-to-date information and will have to make some assumptions or depend upon rumors in order to provide commentary and advice to the reporters. It is vital that the scientific community continue to have rapid access to an authoritative source of results from recent observations.

#### 4. DESIRED CHARACTERISTICS OF A SETI DISSEMINATION SYSTEM

Rapidity, reliability and the ability to target the appropriate audience are the characteristics which any system must have. Initially, speed is

important so that verification may be made as soon as possible. As the confirmation process continues, the rapid distribution of new information to the scientific community will allow us to fulfill our duty of helping to interpret new developments to the public through the channels provided by the popular media.

Reliability is, likewise, of great concern in choosing a system. It is important that a system for collecting and distributing ETI information works the first time. The discoverers must know where to report their first finding and potential verifiers must be aware of, and listening to, the distribution network. To assure reliability, the SETI community must adopt and publicize one single preferred system. Having done that, we must then use or test the system regularly to assure that it will work when called upon. The best way to accomplish this is to adopt a standard system which is already in use for other purposes.

Finally, since the scientific community will play a major role in verifying an ETI signal and interpreting the results of the subsequent analyses of the signal to the public, the information distribution system should be targeted at them.

#### 5. POSSIBLE SYSTEMS

The SETI community could set up a special SETI information system for this purpose only. However, if an existing system can serve the purpose, it is much preferable from a cost and reliability standpoint to tap into it. The following systems for collecting and disseminating information are already in use. While the list is not exhaustive, the first one appears to be an ideal choice to use for SETI.

IAU Central Bureau for Astronomical Telegrams: The International Astronomical Union (IAU) has, since its inception in 1926, run a Central Bureau for Astronomical Telegrams (CBAT) where information on interesting, time-critical astronomical events are collected and distributed to astronomical observatories, laboratories and universities around the world. The International Astronomical Union is composed of adhering organizations for each country which are sanctioned by the those country's governments.

The IAU/CBAT can receive messages by telegram[4] and a number of computer networks. The Bureau, in turn, notifies subscribing organizations either by telegram, dial-up computer connections or by announcement cards sent by air mail. Virtually

every observatory in the world subscribes to this service. Additional subscribers can be accommodated easily. The IAU/CBAT is now experimenting with the use of existing computer networks.

The staff of the IAU/CBAT is experienced in evaluating and handling reports of events where timing is critical. The experience with the recent supernova is an excellent example. Within hours after the discovery of the first naked-eye supernova in nearly 400 years, by an astronomer in Chile, the discovery telegram was in the hands of the IAU/CBAT, followed almost immediately by a confirmation from New Zealand. Within minutes thereafter the first announcement had been sent to the world's observatories. Shortly after that the regularly scheduled observing programs on all telescopes capable of seeing the supernova had been suspended in favor of quick looks at the new object. Within hours of the discovery, detailed spectral observations were begun with the International Ultraviolet Explorer satellite, adding greatly to our knowledge about the early stages of these cataclysmic explosions.

Throughout the next critical weeks, astronomers around the world were kept up to date on the latest observations through the IAU announcements coming from the IAU/CBAT. The system for alerting the world and spreading information worked exceptionally well. The staff of the IAU/CBAT played an important role in keeping the communications flowing and in organizing various ad-hoc communication arrangements adapted to the special needs of the scientists involved in observing the supernova.

Commercial Communication Networks: There are many commercially available communication systems, including commercial telegraph and telex services. While they are rapid, these systems are normally used for rather mundane communications and the receiving machines are not usually located at observatories. Thus, the machines are not checked regularly throughout nights and weekends. A vital request for verification of a suspected ETI signal would stand a good chance of going unread for several hours, or even days. Commercial systems, by themselves, are not sufficient to do the job of collecting, coordinating and distributing observational information.

Existing Computer Networks: While it is true that many scientific establishments and observatories are connected through computer networks, it also true that this network is still

small in comparison to the number of places which need to be alerted. Computer networks can play an important role, but should be considered an adjunct to the prime SETI information system.

Governmental Channels: There are a number of governmental and intergovernmental networks that might be used for disseminating information about a suspected ETI signal. The use of a network which is under the control of a single government might lead to the misguided temptation to try to suppress information. The use of official intergovernmental channels will, from the author's limited experience, be much too slow to be of any practical use. It is clearly more advantageous to use a functioning international network, run by a prestigious international scientific organization, which already has officially sanctioned responsibility for doing similar things.

Popular News Media: The world's news media will clearly play a major role in bringing any ETI story to the public. The problem will be to limit, as best we can, the sensationalism and fantasy which will work its way into the media. The responsible reporters will be counting on the scientific community for help in keeping ETI stories in perspective. To be of help, scientists will have to be kept abreast of developments. A scientific information network will be a vital addition to the normal news channels.

Scientific Literature: We can expect that much information, as well as considerable speculation, will be wrung from the characteristics of an ETI signal which is received. The normal scientific channels for communicating such information will certainly be used to convey such conclusions. The procedure of having such communications reviewed by scientific peers so that it meets accepted standards of scientific research will be a significant part of this process. Peer review provides critical safeguards, but it does slow down the publication of results. We cannot expect the use of scientific literature to be helpful in disseminating information during the early stages of investigating an ETI signal, but after a few months the normal scientific channels of communication will play an important role.

Summary: When an ETI signal is found, all methods of communication will surely be used to inform people about it. However, during the critical early hours and days after the discovery of a suspected ETI signal, rapid and accurate communication among SETI scientists will be

needed. From recent experience it is clear that the IAU/CBAT will be able to fill this role. There are a number of advantages of choosing the IAU/CBAT system for disseminating information about an ETI signal:

1. The staff of the IAU/CBAT have considerable experience and have proven their competence in dealing with similar discoveries in the past.
2. Astronomers will continue to use the system for announcing discoveries of comets, supernova and other time-critical discoveries and events. We can be sure the system will work for ETI because it will continue to be tested and exercised for non-ETI events.
3. Because of their sensitive instruments and interest in observing the sky, astronomers are the most likely to make a serendipitous discovery of an ETI signal. They are also the most likely to use the IAU/CBAT to report their discovery.
4. The IAU/CBAT is set up to employ telegraph, telex and mail services as well as specialized computer networks. In this way, no interested organization need be left out. Even though communication will be slower to some places, the coverage is quite complete around the world.

#### 6. RECOMMENDATIONS AND SUMMARY

1. Dissemination of information is desirable: Receipt of a message from an extraterrestrial civilization is of great importance for the entire world. News of such an event should be spread to the inhabitants of our planet as rapidly and accurately as possible. All SETI scientists should publicly subscribe to this goal. The scientific community has a major responsibility to help in this process, but scientists must have a means of keeping up with developments.

2. Target first announcements to scientists: Because of the requirement for confirming a suspected signal with a second telescope before accepting its reality, the first announcement of a suspected ETI signal must be done rapidly and must be directed to the scientific community. Follow-on distribution of information must also be targeted at scientists.

3. Use IAU telegram system to distribute scientific information: The International Astronomical Union Central Bureau for Astronomical Telegrams meets all the requirements for disseminating announcements and preliminary information about ETI signals. It is a well established system of rapid communication targeted at astronomers, who

are the most likely to be able to make confirming observations of a suspected signal. Moreover, the system is continually being tested by normal astronomical discoveries. The SETI community should adopt the Central Bureau for Astronomical Telegrams of the International Astronomical Union as the official mechanism for reporting and disseminating SETI information.

4. Provide accurate information for news media: the news media of the world will undoubtedly see that announcements of an ETI signal will reach every country. There will be a great temptation to sensationalize any such event. It is the responsibility of the SETI science community to provide factual information to the press. The IAU/CBAT can help all scientists to stay up-to-date with the latest observations and developments both during and after the process of confirming the extraterrestrial character of suspected ETI signal.

We all agree that receipt of a suspected ETI signal would have a profound impact on our civilization. As soon as word of such a signal starts to circulate there will be tremendous interest and considerable confusion. One can only hope that a signal is confirmed without doubt before the word is circulated widely. However, after an ETI origin for the signal is confirmed, the process of monitoring and analyzing the signal will, by default, become a very public process. As scientists, those of us engaged in SETI will be drawn into the process of informing the public. If we can prepare for this event by ensuring that we have a rapid and accurate means of sharing information, we will be better able to meet the challenge that a verified ETI signal will bring.

#### REFERENCES

1. T. M. Hesburgh, foreword to *The Search for Extraterrestrial Intelligence*. P. Morrison, J. Billingham and J. Wolfe, eds., NASA SP-419 (1977).
2. P. Morrison, *ibid.*, p. 3.
3. P. B. Boyce, paper #IAA-86-473 given at 37th IAF Congress, Innsbruck (1986) (included in this special issue).
4. D. W. E. Green, *Sky and Telescope* 74, p. 421 (1987).